

Spatiotemporal coherence of MHT in the Atlantic Ocean

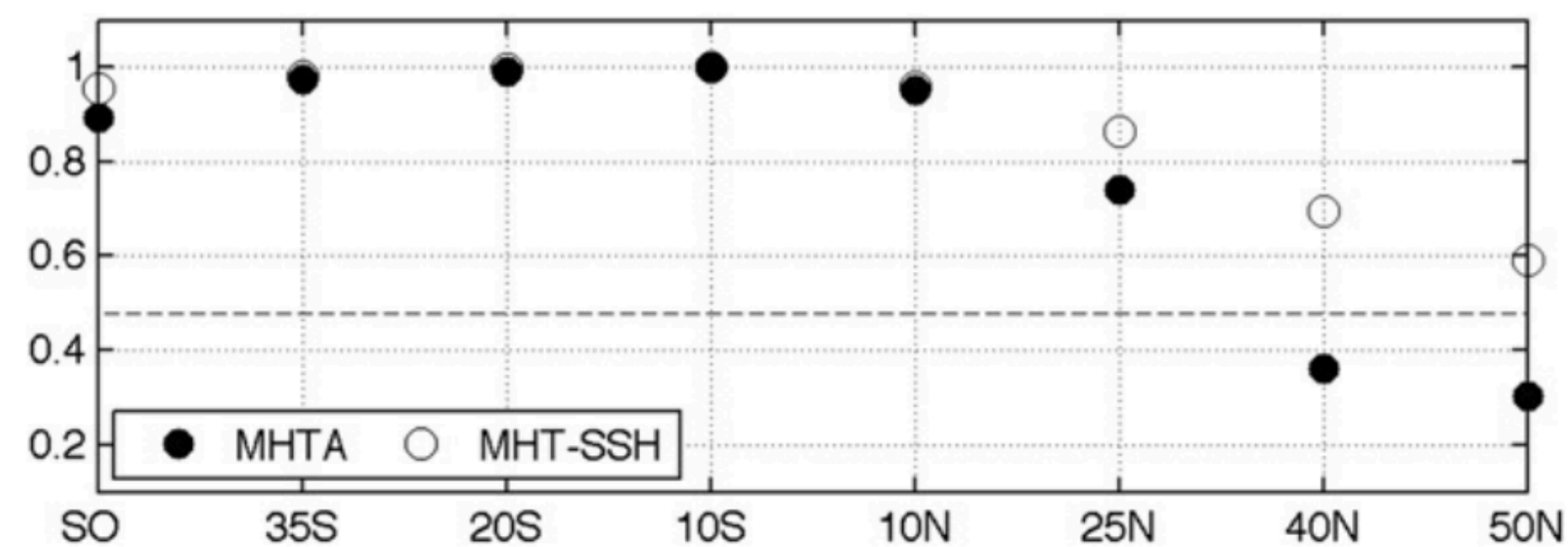
1. Event : breakdown/split of coherence around 2009

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Literature says:

Kelly et al. (2014):

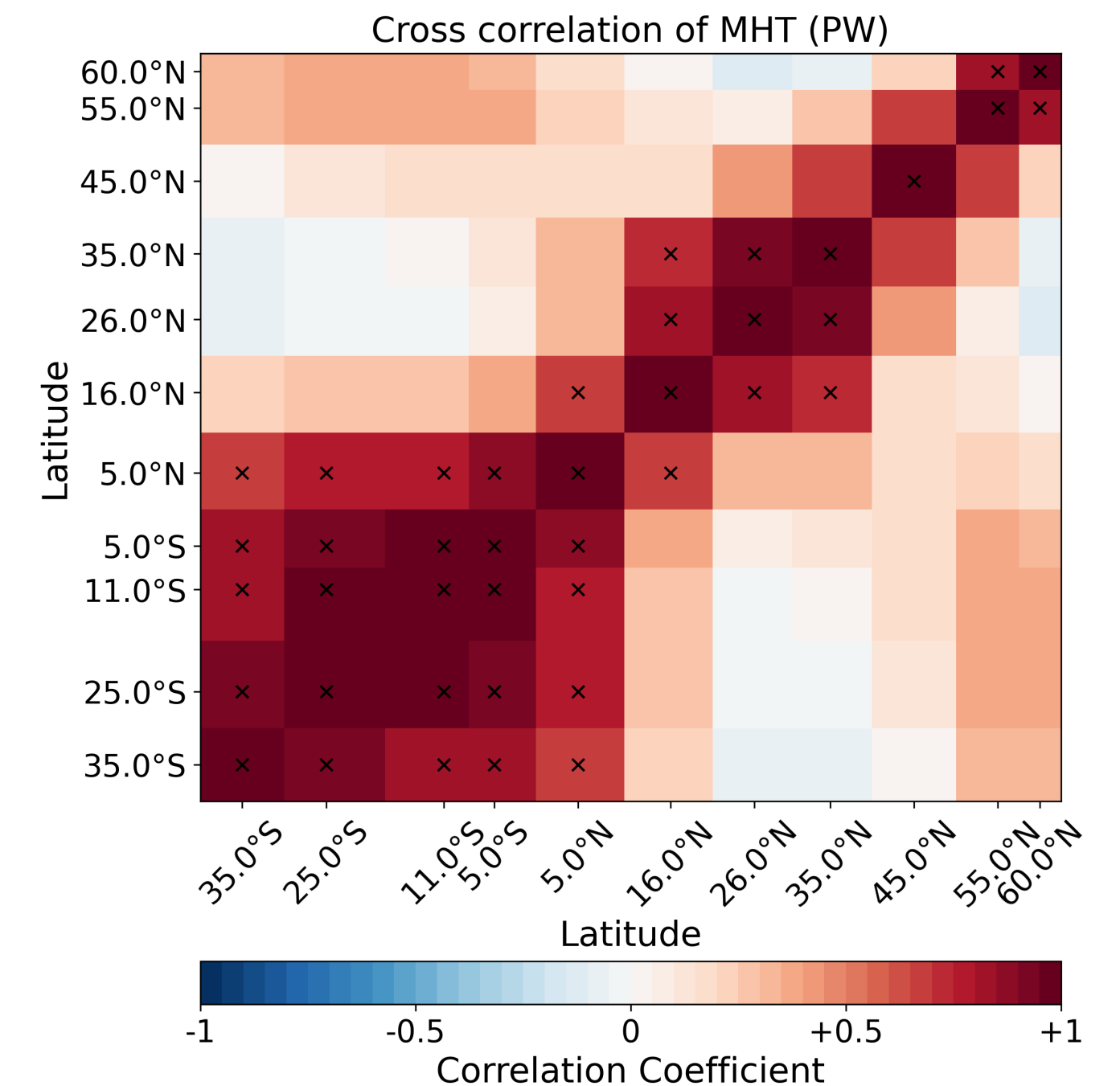
- Coherence south of 40°N up to SO
- Timeperiod: 1993-2011



Calafat et al. (2025):

- 2 Covarying blocks:
- 35-16°N and 5°N-35°S
- Timeperiod: 2004-2020

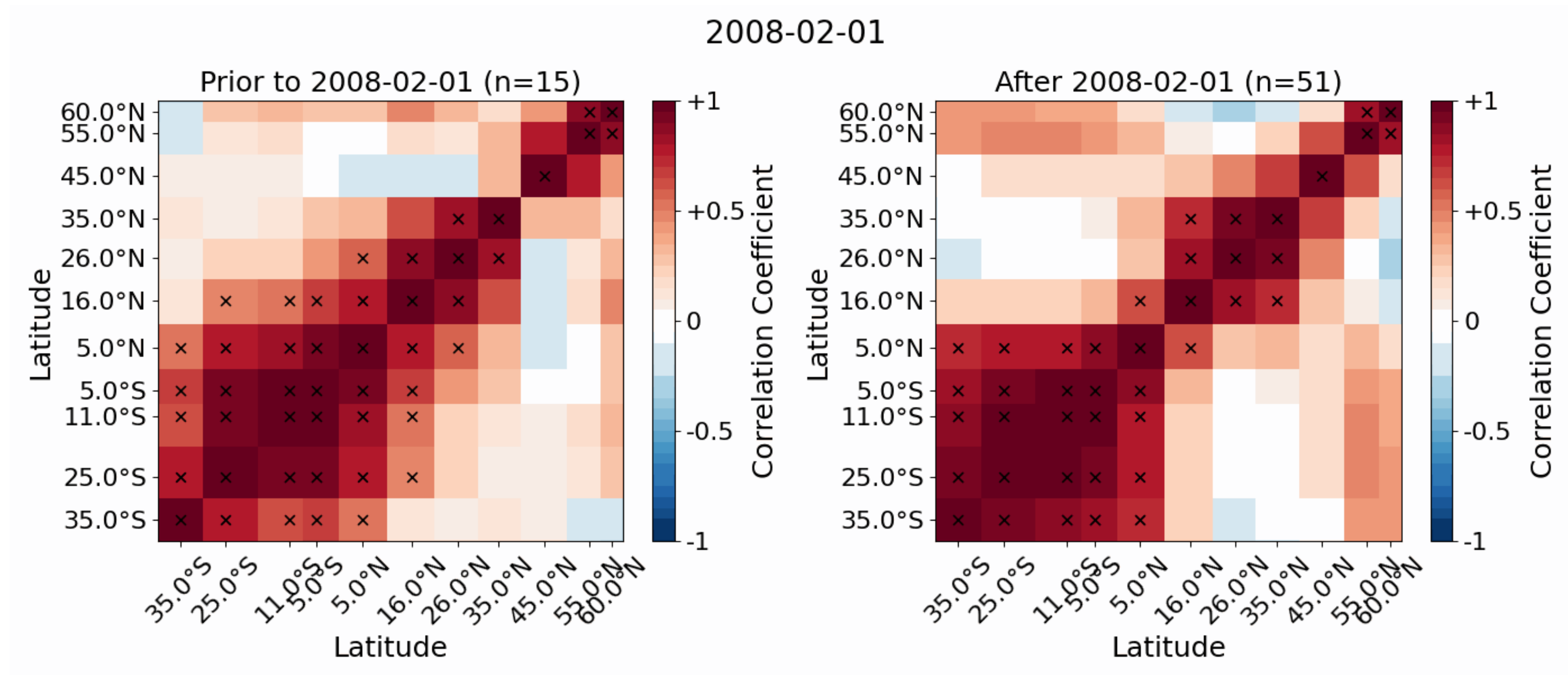
With this data I get:



When is this split happening and what is causing it?

Idea:

- split timeseries into before and after split year
 - plot gif sliding the split year through the timeseries
- This led me to distinguish 3 time periods

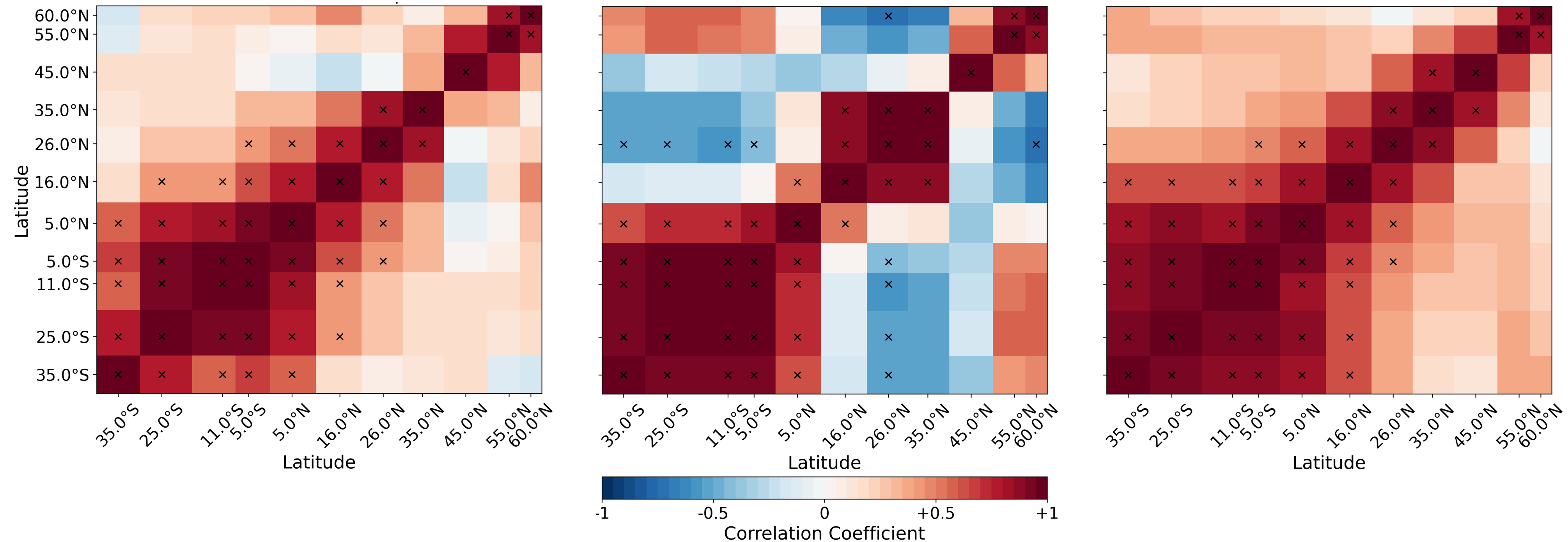


Split whole timeseries into 3 periods:

Pre 2009

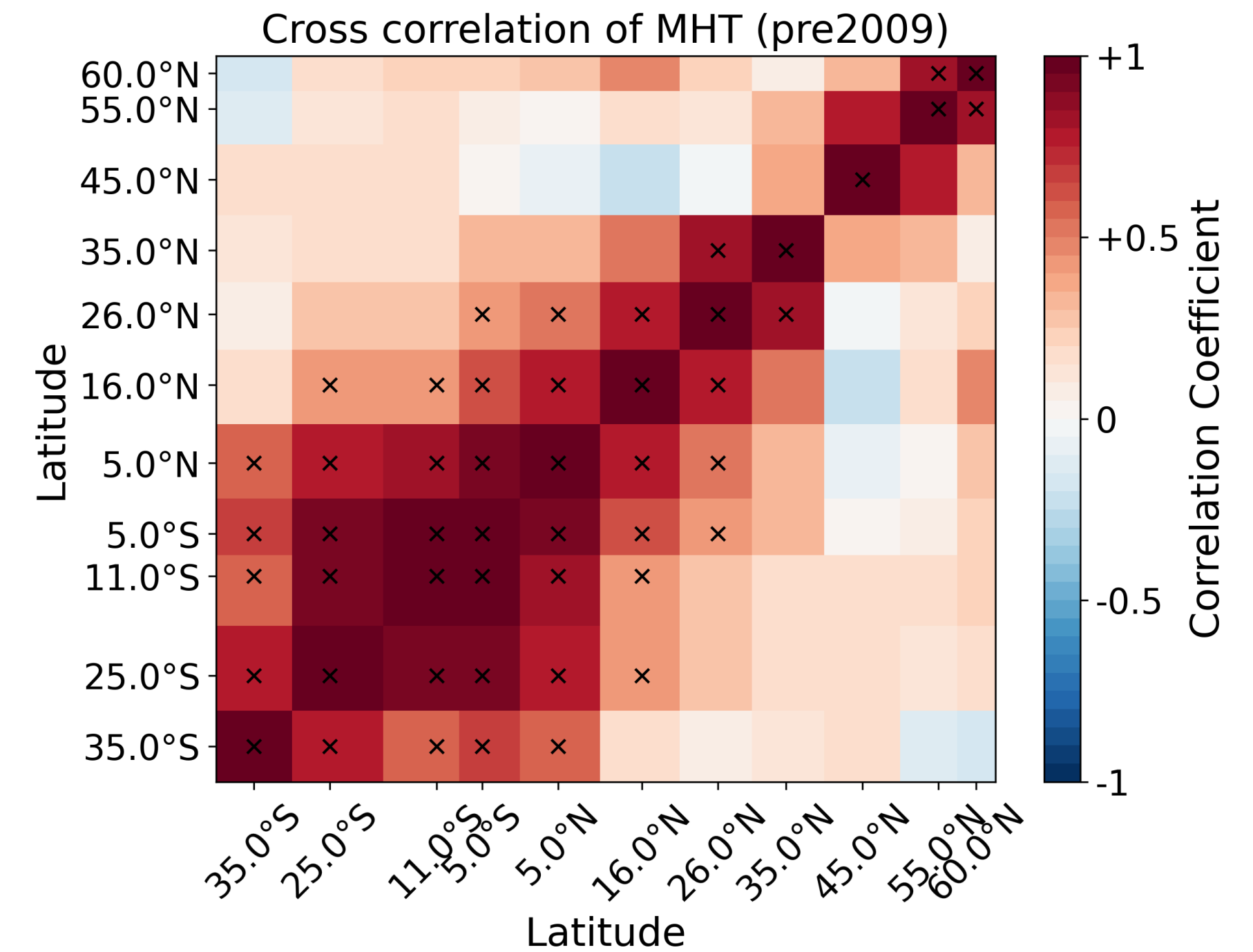
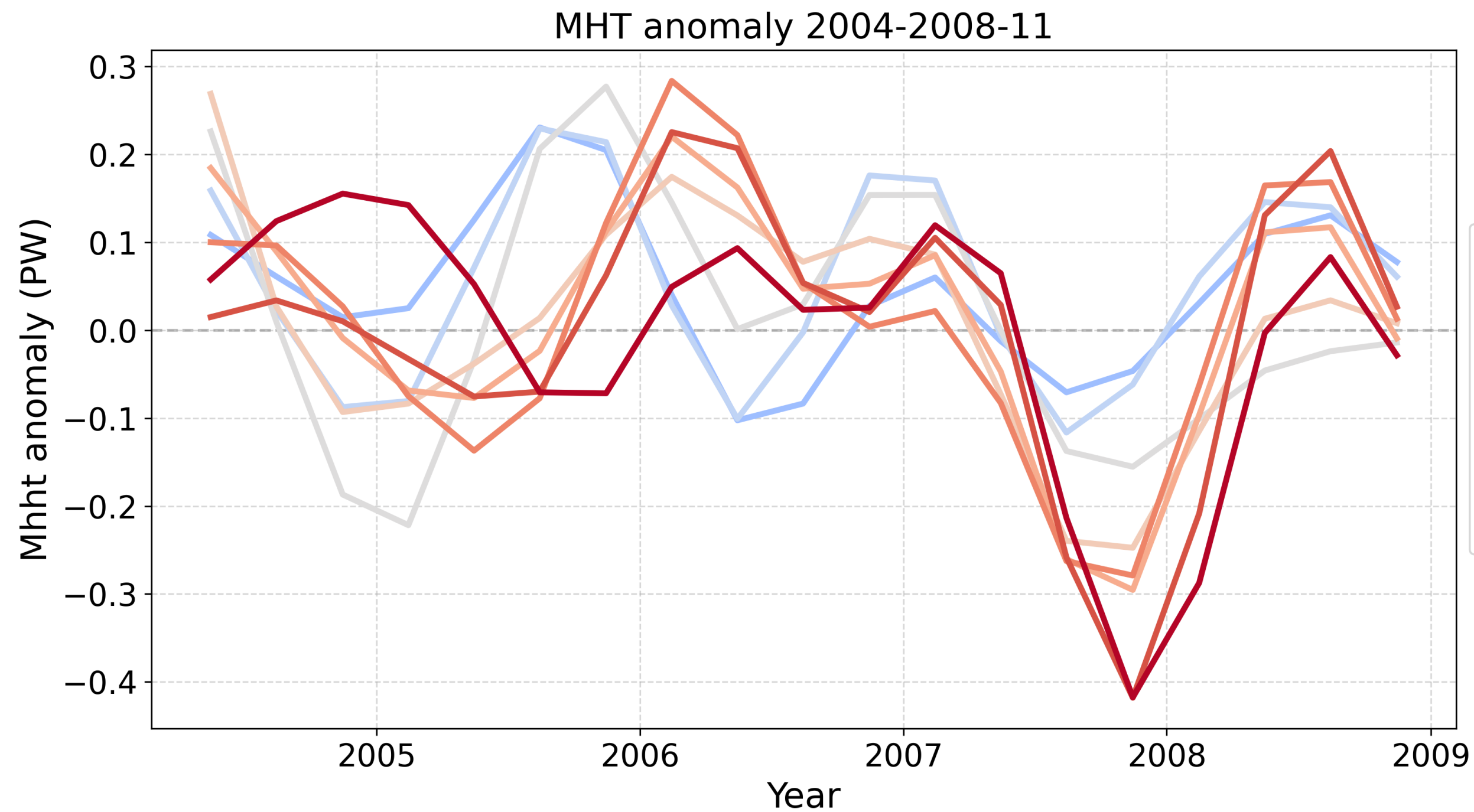
2009-2013

Post 2013



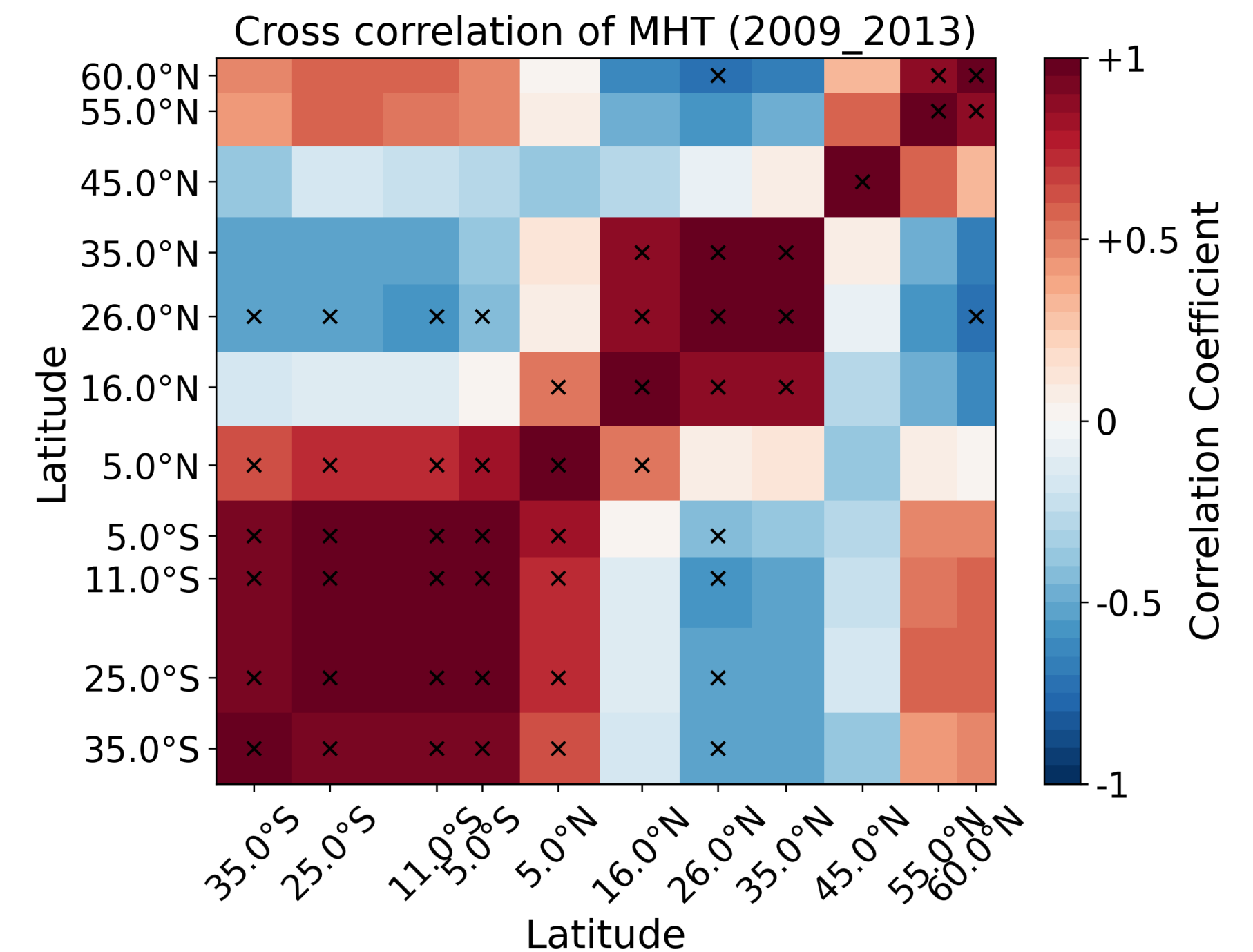
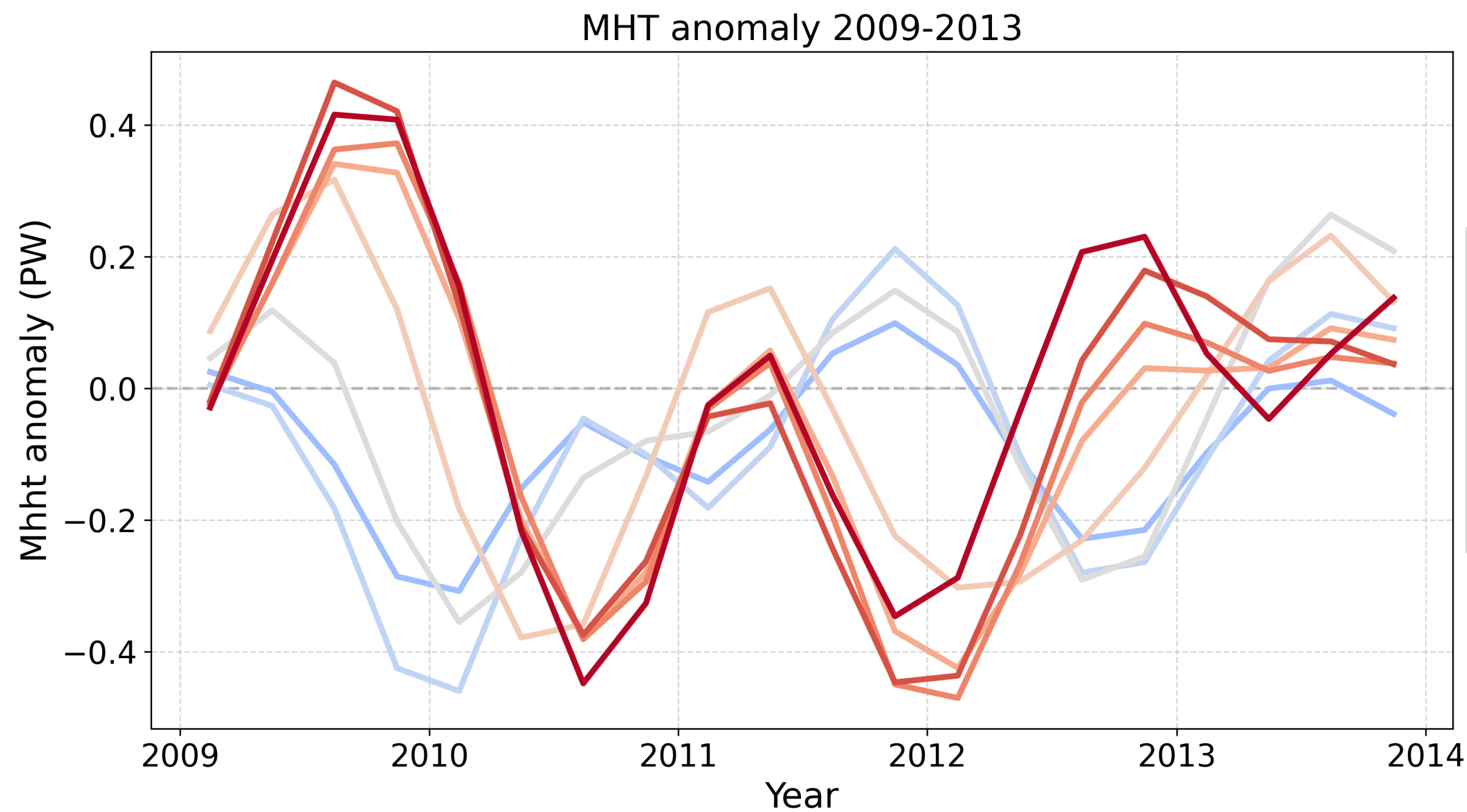
- Before 2009: high correlation throughout Atlantic south of 35°N
- Then pattern shifts and moves southwards, splits into two boxes around 2009
- Two covarying boxes stays until 2013

MHT Coherence before 2009



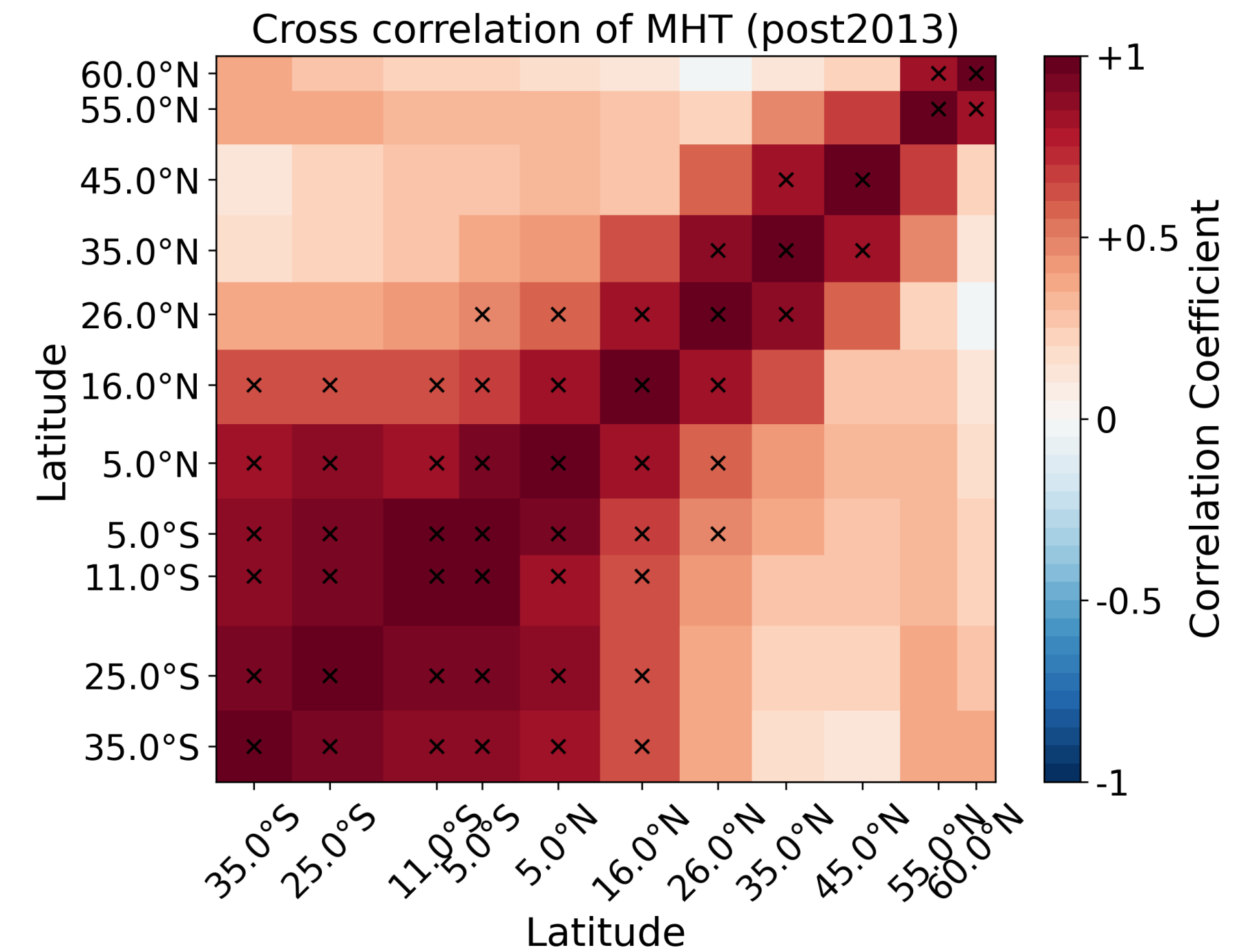
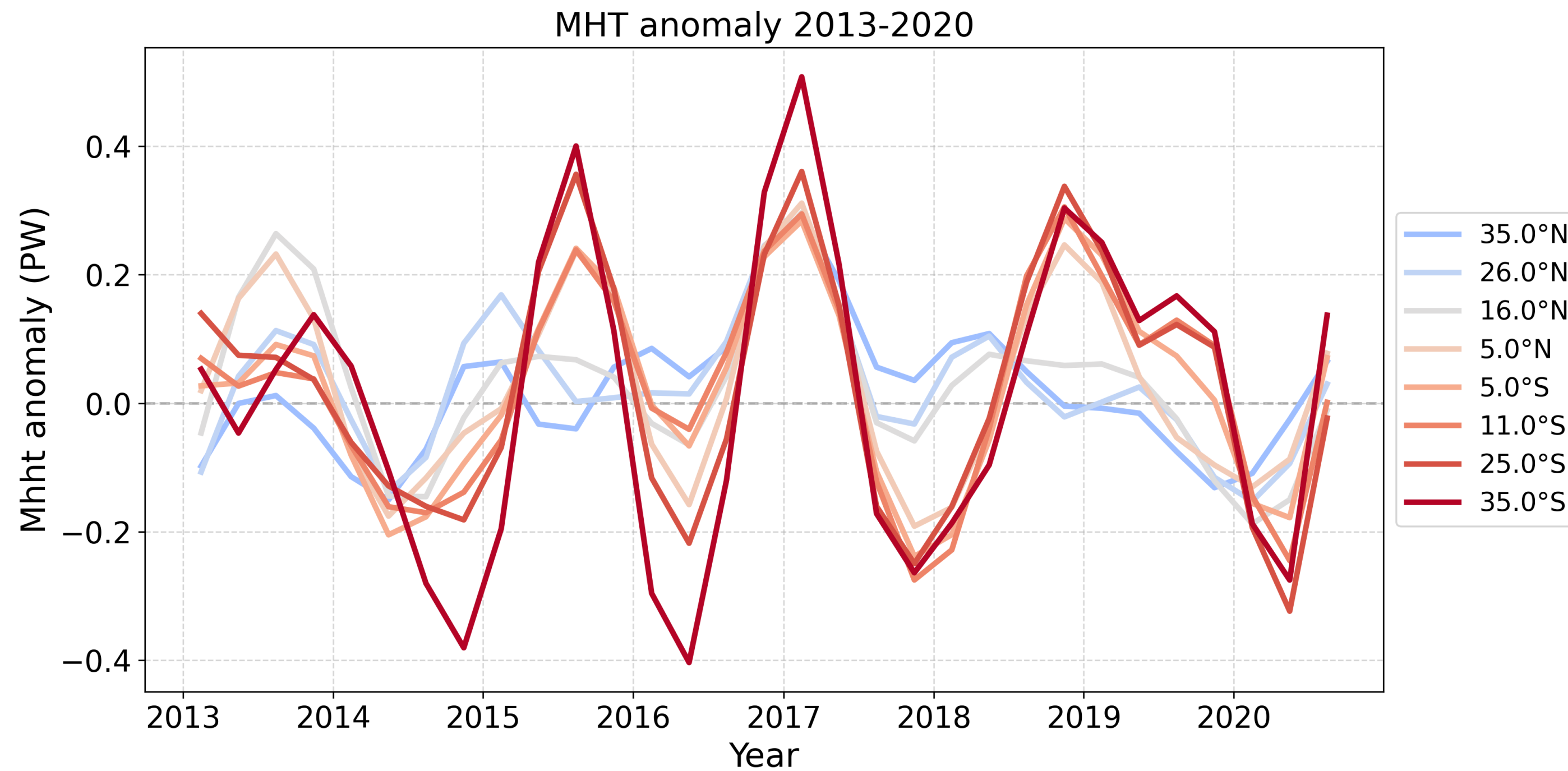
- Mostly **in phase signal** and **amplitude**, except for mid/end 2007
- Mean pairwise correlation for block **35°N-35°S** : 0.71

MHT Coherence / correlation break 2009 - 2013



- Here we see signal goes from covarying and **in phase** to **anti correlated**
- split into 2 covarying boxes
- High amplitudes for southern signal
- Mean pairwise correlation for block **35°N-35°S**: 0.54

MHT Coherence after 2013



- Again mostly **in phase signal** and **amplitude**, northern latitudes tend to have lower amplitudes
- Mean pairwise correlation for block **35°N-35°S** : 0.79

Next steps / open questions:

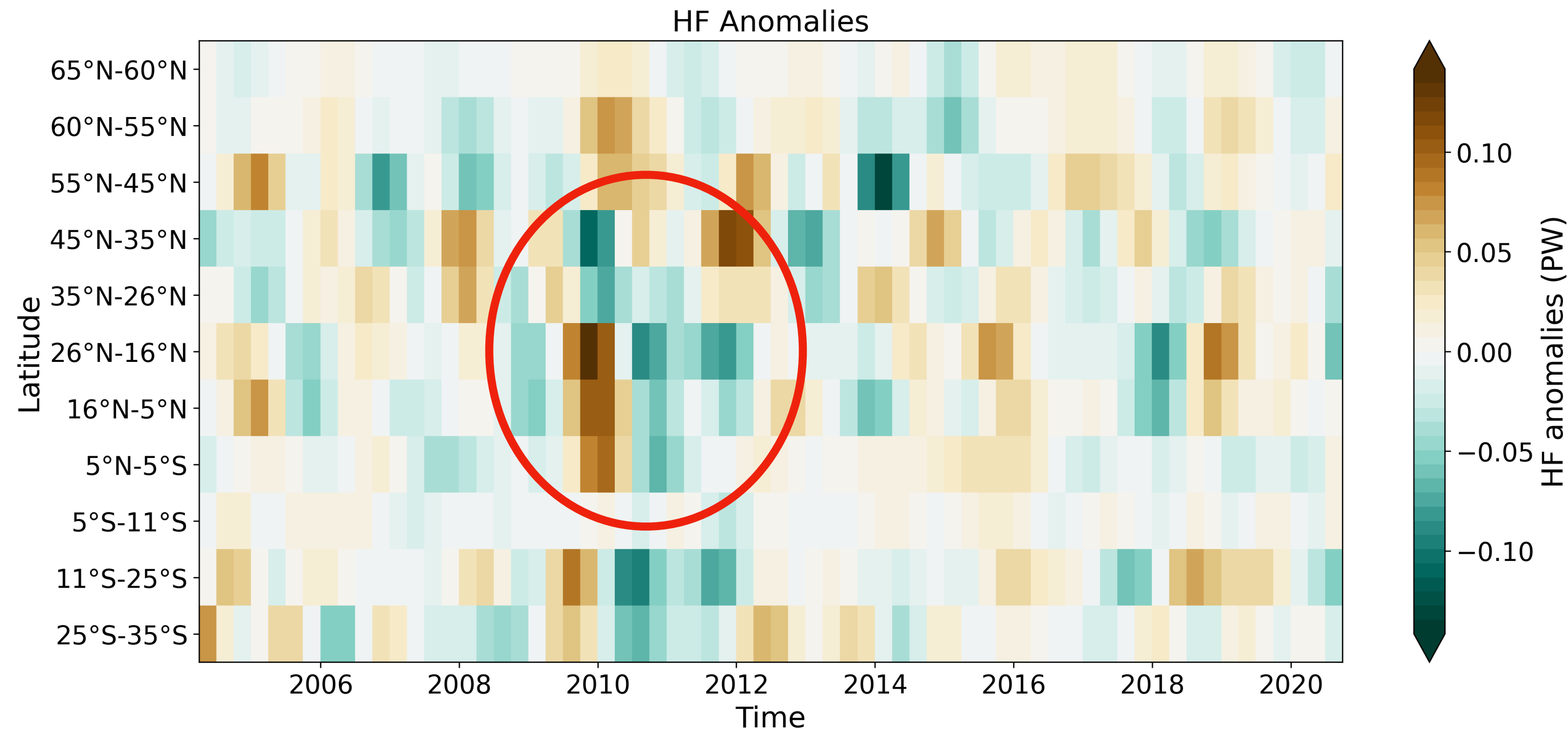
Find some propagation in the signal, do a max lag correlation plot ...

What caused the coherence break and split into two boxes?

- 2008/2009 record low NAO → AMOC & MHT drop
- Connect NAO - air sea heat flux F_S - MHT

Why is the break around 5-16°N?

HF anomalies



- Around 2010 very strong positive HF anomaly (strong uptake) around equator
 - After NAO event in 2008/2009
- Followed by negative anomalies (heat released) middle of 2010-2011
- Coherence break around these latitudes, 1 year before (beginning of 2009)